

RAJNANDINI PATIL

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GitHub: <https://github.com/Rajnandini15122006>

LeetCode: <https://leetcode.com/u/rajnandini15122006> | **CodeChef:** https://www.codechef.com/users/free_lily_41

PROFESSIONAL SUMMARY

Pre-final year AI & DS student at VIT Pune (CGPA 9.65) with strong foundations in data structures, algorithms, and system design. Built production-ready systems using Python, FastAPI, and LangChain. Published researcher at CNC 2025. Solved 740+ LeetCode problems with Contest Rating 1490+. Active in competitive programming and 25+ national hackathons.

TECHNICAL SKILLS:

Languages: Python, C++, JavaScript, SQL

Backend: FastAPI, Node.js, REST APIs

AI/ML: Deep Learning, LLMs, LangChain, RAG, FAISS, Graph Neural Networks, Scikit-learn

Databases: PostgreSQL, MongoDB, Redis, Firebase

Tools: Git, GitHub, Docker, Linux, VS Code

Libraries: NumPy, Pandas, BeautifulSoup, Matplotlib, Seaborn

TECHNICAL PROJECTS:

Distributed Search Engine (Mini Google) | Python, FastAPI, React, PostgreSQL, Redis, Docker

[GitHub](#)

- Built multi-threaded web crawler using Python threading for concurrent URL crawling and indexing
- Implemented TF-IDF ranking algorithm from scratch without ML libraries for relevance-based search
- Designed inverted index in Redis for sub-millisecond document lookup across large page collections
- Built React frontend with real-time search and debouncing for optimized performance
- Deployed full stack using Docker and docker-compose

AML-Shield – Anti Money Laundering Detection | Python, GCN, GAT, FastAPI, Docker

[GitHub](#)

- Built fraud detection system on Elliptic Bitcoin Dataset with 200,000+ transactions and 49 features
- Implemented Graph Convolutional Networks (GCN) and Graph Attention Networks (GAT) to model transaction networks and detect illicit patterns
- Combined supervised classification with Isolation Forest anomaly detection for real-time risk scoring
- Developed FastAPI backend with interactive network visualization dashboard

PROFESSIONAL EXPERIENCE:

NLP/LLM Intern – HCL Technologies

Sep 2025 – Present

- Contributed to RAG-RuleSync LLM pipeline extracting structured rules from unstructured technical documents
- Worked on JSON schema design and structured extraction pipeline using LangChain and Groq API
- Collaborated with senior engineers on RAG architecture and LLaMA 4 integration

AI & Decentralized Systems Intern – BlockseBlock

July 2025 – Sep 2025

- Developed AI and blockchain-based workflows using Python and decentralized identity frameworks
- Built automated identity verification pipelines for decentralized applications
- Supported smart contract-aligned automation workflows

Machine Learning Intern – Codveda Technologies

Sep 2025 – Nov 2025

- Built and evaluated ML models for predictive analytics using Python, NumPy, Pandas, Scikit-learn
- Performed data preprocessing, feature engineering, and dataset preparation for model training
- Improved model performance through hyperparameter tuning and cross-validation techniques

Data Science & Machine Learning Intern – Infotact Solutions

Jan 2026 – Feb 2026

- Built predictive analytics models on structured datasets using Python and Scikit-learn
- Developed data preprocessing and model evaluation pipelines for real-world data

ACHIEVEMENTS:

- Published two research papers at International Conference on Advances in Communication, Network, and Computing (CNC 2025), Bengaluru
- Selected contributor – GirlScript Summer of Code (GSSoC) open-source program
- Selected participant – Winter of Code 2.0 open-source program
- Finalist in multiple national hackathons across 25+ competitions
- Contributed to OWASP open-source security initiatives

COMPETITIVE PROGRAMMING:

- Solved 740+ DSA problems on LeetCode (Contest Rating 1490+)
- Active participant on Codeforces and CodeChef (Contest Rating 1334+)
- Strong in graphs, dynamic programming, binary search, trees, sliding window, bit manipulation

EDUCATION:

Vishwakarma Institute of Technology, Pune

2024-2028

Bachelor of Technology – Artificial Intelligence and Data Science | 2024 – 2028

Relevant Coursework: DSA, Operating Systems, DBMS, OOPS, Machine Learning, Deep Learning, Computer Networks, Probability & Statistics, Linear Algebra, Artificial Intelligence

CGPA: **9.65** / 10